AI itself is not a frontend ecosystem, but it has become a foundational layer within the frontend development ecosystem. Rather than being a standalone ecosystem, AI acts as a transformative set of technologies and tools that integrate with and enhance existing frontend workflows, frameworks, and platforms. AI-driven tools and agents now automate code generation, optimize design and layout, improve accessibility, and personalize user experiences, all within the broader frontend development environment. These technologies are embedded in the ecosystem, working alongside frameworks like React, design tools such as Figma, and CI/CD pipelines to streamline development, boost productivity, and enable new capabilities.

A portion of this book is dedicated to practical implementation, preparing candidates for coding challenges commonly faced in frontend interviews. Strategies for navigating React coding interviews are provided, along with mock exercises that simulate real-world scenarios, such as building a star rating component, an image slider, and a weather app. These hands-on exercises reinforce technical skills and help candidates gain confidence in their ability to solve problems under pressure.

Upon completing the basic star rating component, an interviewer can increase the challenge with the following suggestions:

- Implement a feature that allows users to click on a star to set a rating and then click again to clear it.
- Modify the implementation to support half-star ratings.
- Add animation to the star-filling process when a rating is selected.
- Make the star rating component fully accessible using ARIA attributes.

These advanced challenges, explored in chapter 8, encourage candidates to consider interactive features, visual polish, user experience enhancements, and accessibility—all crucial aspects of modern frontend development that interviewers often look for in top candidates.

Recognizing the increasing importance of system design in frontend roles, modern interviews may include a dedicated section focused on system design questions. This segment equips UI engineers with the skills and knowledge necessary to address complex challenges and collaborate effectively with backend teams. Candidates are expected to engage in discussions around key